

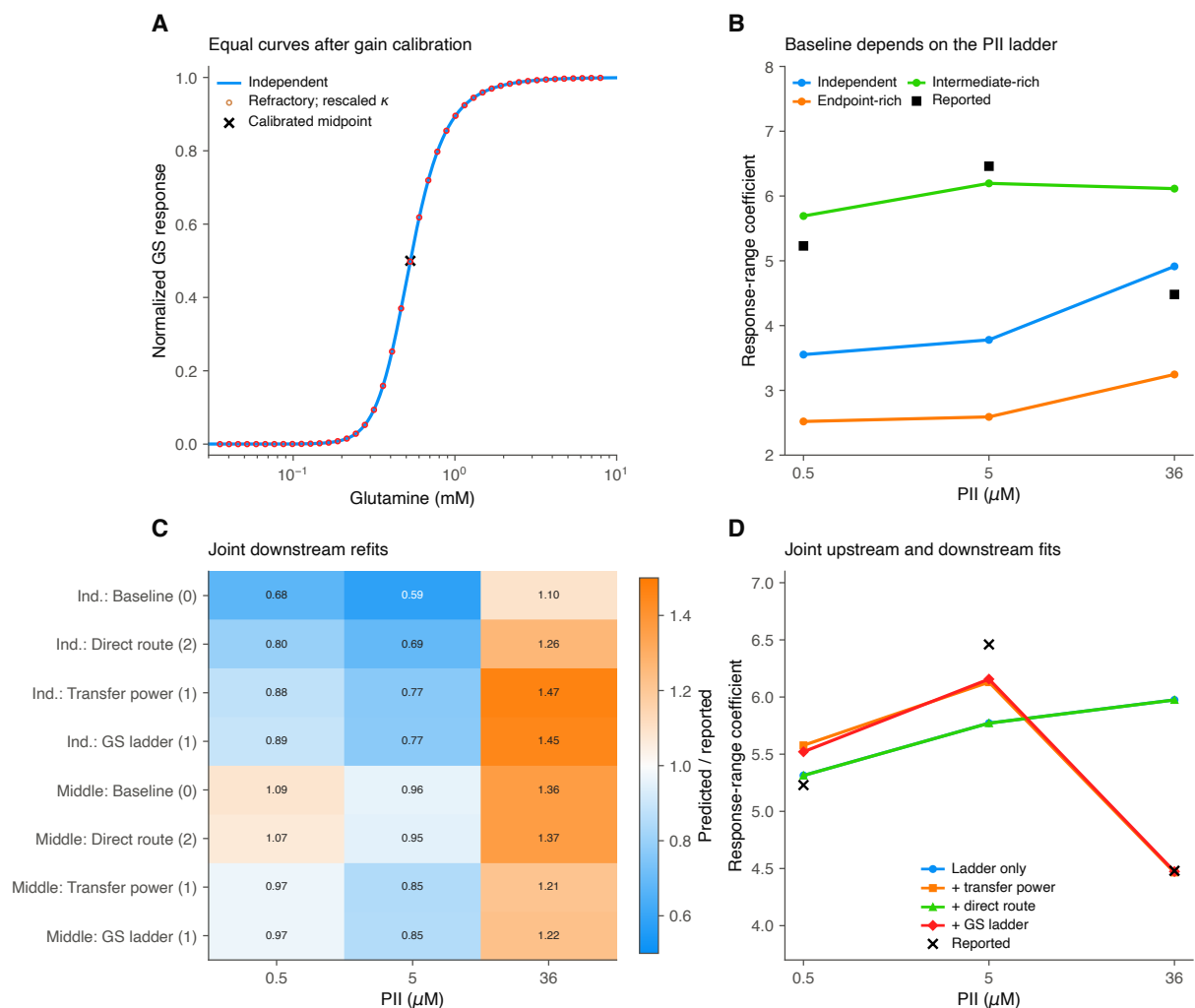


Figure S15: Matched calibration separates observational equivalence from model-dependent discrepancy. (A) Identical independent-population and refractory-population cascade curves after gain rescaling, shown at the middle-condition midpoint. Curves are predictions; the cross is a calibration target. (B) Baseline response-range predictions from three exact mean-matched ladders and reported summaries. (C) Predicted-to-reported ratios after joint fitting for fixed independent or intermediate-enriched PII ladders. Parentheses give the number of added shared downstream parameters; all models also use three condition-specific midpoint gains. (D) Joint comparisons allowing one additional shared upstream ladder parameter. The transfer-power model is a phenomenological benchmark. No summary errors are available for likelihood-based selection. Smooth lines connect the three assay conditions and do not establish a concentration law.